

AMIT KUMAR

Senior Hardware Design & Product Development Engineer

Industrial | Energy | IoT | Telecommunications

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PROFESSIONAL SUMMARY

Senior Hardware Design & Product Development Engineer with 15+ years of experience across Industrial, Energy, IoT, and IT domains. Extensive expertise in end-to-end hardware product development, covering requirements definition, system architecture, detailed design, prototyping, bring-up, validation, qualification, certification, and final delivery. Strong background in high-speed digital hardware design, power electronics, EMI/EMC compliance, hardware-software integration, and IoT product development. Proven ability to lead engineering teams, architect complex hardware systems, and deliver production-ready, high-reliability products.

CORE SKILLS

Hardware Design & Architecture

End-to-end product development • System architecture • Design specifications • Component selection • Schematic design & review • Power estimation & budgeting • Board bring-up • Hardware validation & debugging • Failure analysis • Battery charging circuits • Sensor-based designs

High-Speed & Digital Interfaces

USB • DDR3 / LPDDR4 • Ethernet • LCD • SD / eMMC • CAN • I2C • SPI • UART • RS-232 • RS-422

Wireless & Connectivity

LTE (3G/4G) • Wi-Fi • Bluetooth / BLE • ZigBee • Sub-GHz RF • IoT gateways

Processors & Platforms

TI DSP • NXP i.MX6 / i.MX8 / i.MX9 • Xilinx FPGA • PIC • NXP & STMicroelectronics MCUs

Simulation & Analysis

LTspice • OrCAD PSpice • DC-DC design • AC/DC analysis • MATLAB signal analysis

PCB & Manufacturing

PCB layout review • Layer stack-up definition • Placement & routing guidelines • DFM / DFT • Manufacturing handoff

Compliance & Qualification

EMI/EMC (ESD, EFT, Surge, RE, CE, RS, CS) • IEC 61000 • IEC 61326 • Environmental & reliability testing • Airborne & Military-grade product exposure

Leadership & Collaboration

Team leadership (4–5 engineers) • Hardware-software integration • Cross-functional collaboration • Product lifecycle management

PROFESSIONAL EXPERIENCE

Bosch Global Software Technologies Pvt. Ltd. — Hardware Architect,
Oct 2022 – Present

Tata Consultancy Services (TCS), Bangalore — Senior Hardware Engineer
Feb 2019 – Oct 2022

Robert Bosch Engineering & Business Solutions Ltd. — Senior Hardware Engineer
Feb 2015 – Jan 2019

Accord Software & Systems Pvt. Ltd., Bangalore — Senior Systems Engineer
Jul 2013 – Feb 2015

Accord Software & Systems Pvt. Ltd., Bangalore — Systems Engineer
Sep 2010 – Jun 2013

KEY PROJECTS

Telecommunication Gateway – Industrial IoT

- Designed an **industrial communication gateway** supporting **Ethernet, USB, SD Card, CAN, RS-485, and UART** interfaces.
- Architected a **secure communication pipeline** for industrial networks and cloud platforms.
- Designed **power supply architecture**, protection circuits, and **EMI/EMC-compliant layouts**.
- Performed **hardware bring-up, validation, qualification, and compliance testing**.

In-Meter Gateway – Sub-GHz + LTE

- Developed an **embedded in-meter IoT gateway** with **Sub-GHz RF and LTE connectivity**.
- Enabled **direct meter-to-cloud communication**, eliminating external concentrators.
- Designed **compact RF front-end**, integrated **LTE module, SIM interface, power sequencing**, and **ESD/surge protection**.
- Conducted **antenna tuning, RF optimization**, and **EMI/EMC compliance testing**.

NIC Card – Smart Metering

- Designed a **Sub-GHz RF NIC card** for meter data acquisition.
- Owned **hardware architecture, component selection, schematic design**, and **PCB layout review**.
- Optimized **RF performance** and ensured **regulatory compliance**.

DCU Gateway – Smart Metering

- Developed a **Data Concentrator Unit (DCU)** for **Sub-GHz to 4G/Wi-Fi to Cloud** data transfer.
- Implemented **USB interface** for diagnostics and service operations.
- Executed **hardware validation, environmental testing, and field performance improvements.**

Non-Intrusive Load Monitoring (NILM)

- Responsible for **hardware architecture, schematic design, and PCB layout review.**
 - Performed **design verification using LTspice simulations.**
 - Conducted **hardware validation, EMI/EMC testing, and environmental qualification.**
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EDUCATION

- **Bachelor of Engineering – Electronics & Communication**
Basavakalyan Engineering College, VTU, Karnataka, **Score:** 79.64%
 - **12th Standard** – M.J.K College, Bettiah, Bihar, **Score:** 64.44%
 - **10th Standard** – Yugal Prasad High School, Bhaisahi, Bihar, **Score:** 70.43%
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ADDITIONAL INFORMATION

- **Languages:** English, Hindi, Bengali
- **Interests:** Cooking, creating cooking videos, reading